

ActInf Livestream #022.1 ~ Psychophysical identity and free energy

Livestream | Psychophysical identity, free energy | 1:50:30

okay hello

and welcome everyone to actin flap

this is the actin flap live stream

and i'm pretty sure we're live yes we
are

it's number 22.1 on may 18

2021 and this is going to be really
great discussion so i'm looking forward
to it and thanks again everyone for
joining live welcome to

actin flap we are a participatory online
lab

that is communicating learning and
practicing applied active inference

you can find us at the links that are
here on this page

this is recorded and an archived live
stream so please provide us with
feedback

so that we can improve our work all
backgrounds and perspectives are welcome
here and we'll be following

good live stream etiquette welcome scott
and

raising our hands to ask questions and
hear from everybody because this is
really going to be a special discussion
we're here in paper number 22 and on may
18th today

and next week on the 25th we're going to
be having two discussions
on the paper psychophysical identity and
free energy

by alex kiefer so thanks a lot alex for
joining us live and engaging with the

dot
zero video because we gained a lot as a
lab
working through that and so we're ready
to take it to the next level through
conversation
with you thank you thank you and today
we're gonna be
learning and discussing the papers link
is here
and the video is just us unpacking and
exploring
raising questions and we're going to be
also taking notes so that in the dot 2
we can be following up we're going to
start with some introductions
just to introduce ourselves and also
maybe to raise any thoughts
or questions that we had at the outset
about the paper so
everyone can feel free to write down
points or write down questions so they
don't uh forget them
and we're pretty much just gonna jump
around to any
topic or question that drives us along
and we'll flow with it as people join
and leave
and there will be a follow-up discussion
22 next week
that brings us to the introductions and
the warm-ups
so i'll start and we'll pass it around
and then end up with alex
who will provide i think a short
introduction to the paper
and then we'll continue the discussion
from there

so i'm daniel i'm a postdoc in
california
and something i'm excited about today
was
something we were just speaking about
right before was learning about a
transparent code
and understanding what that might mean
i'll pass it to
scott

hello i'm scott david i'm at the
university of washington applied physics
lab

where i run a program called the
information risk and synthetic
intelligence research initiative
and i'm delighted to be here today and
i'll pass it to stephen

hello i'm stephen i'm based in toronto
i'm doing a practice-based phd
at canterbury christchurch and um i'm
really excited about
the psychophysical areas being brought
in with this
process with energy so i'm intrigued
because that's

connects to some of the research i do
with communities in their landscape
and how to understand that so thank you
and i'm going to pass it over to alex
thanks hi everyone i'm alex watkins
i'm a researcher in systems management
school in moscow russia
and also co-organizer of acting club
and i pass it to dean

hi i'm dean i'm up here in calgary
canada

i'm retired and i just try to keep up

with everybody here

i'll pass it to our special guest today

alex

hey dave um no dave is left sorry okay

continue okay i'll excuse me

hey thanks uh yeah i'm alex kiefer um

i'm a researcher in philosophy and

cognitive science and i do some

modeling as well um i'm i have an

affiliation with monash university

um i'm currently also a lot of my time

is devoted to working

for uh nested minds solutions which is

um

an enterprise uh started by maxwell

ramstad and some other people

uh just recently uh we've already got a

lot of work

under our belt um so i'm really happy to

be here um

thanks thanks for having me this is like

i think gonna be a really fun discussion

um cool maybe just

while the other participants are

preparing a thought

or a question to lead off alex just how

did you

come to writing this single author paper

what yeah in your path led you to write

it this way

yeah sure um so uh so this is an idea

that

i think first sort of uh came up for me

in

2015 or so um

so i've been fascinated by this formal

analogy between

optimization and uh processes and

thermodynamics and physics
um for a long time since well not that
long maybe 2012 2013 when i started
getting into the neural net literature
on
helmholtz machines and things like that
and um
so i think this this analogy uh between
physics and optimization has been
tempting
for a lot of people to want to collapse
these right and this is something that
people like um
my friend mel andrews constantly warned
against is like just taking the physical
informational description of the system
to be uh the same
as as its physical description but i
thought that there was
a compelling argument to be made here so
i wanted to actually articulate not
necessarily endorse
right i think the way i couch things in
the paper is a bit cautious on this but
i wanted to articulate
what is the most sort of the simplest
most kind of perhaps radical perspective
one could take on this
the relationship between the cognitive
dynamics and the physical dynamics
um that you might be able to defend
right from a sort of free energy
principle-ish perspective
um so i wanted to to um just
lay that that one way of understanding
this relationship out that was
that was as clear as possible um i think
in the process i ended up writing

a paper that many people find to be quite complicated so this was supposed to be an exposition of a simple idea so maybe we can see if we can get back to that

awesome so anyone can raise their hand and we'll just start but i really liked how you said articulate but not endorse you've put out an artifact for us to interact with and to explore and you have framed the psychophysical identity thesis as a hypothesis so it's something that evidence can now be used to support or to disprove and that's sort of um

a nice piece of work that now obviously we have started the discussion around so we'll go to stephen and then scott and then anyone else with a raised hand

one thing i like about what you're doing here is you're sort of bringing it into the domains where the action is happening it's a bit like most of the physics around thermodynamics is really often used in chemistry and a lot like niels bohr was actually a chemist wasn't a physicist per se he used physics so there's an interesting thing about how these areas get transposed over you know and may be layered again to biochemistry which maybe is trying to look at the

mechanisms that sit on that
and xyz so i think that's quite relevant
to
some of the questions we've been
wrestling with with what happens when
things start to transit
transition to the ontology like and how
how can you keep a handle on that when
you've got people from different
backgrounds so i i think that's quite
interesting awesome thank you stephen
scott and then anyone else with a raised
hand
yeah i'd like to just ask about your
having written the paper and now
contemplated it discussed it with folks
what's your impression of the role
of the observer of difference
of gaps of differences
in identifying sources
of free energy
um exploitable sources let me let me
back up
one i did a paper in uh presentation in
um
mit in 2013 called entropy
and i asserted that carnot's equation
the hot and cold differentials that are
reflected there are the same as
arbitrage equations
in markets because you and i asserted
that heat engines were produced by the
human mind
observing an attribute because it didn't
matter that we thought we had a flow of
caloric
and it was flowing downhill we actually
the notion of flow

was what we asserted in order to convert
random heat random motion of particles
which we really only use for warmth
being able to put that into a cylinder a
combustion chamber and then perform work
took thousands of years i wonder if you
could comment on
it feels like we're using information
like the cavemen
as for warmth and not yet putting it in
a combustion chamber
and the combustion is it the fact that
the processes
where differentials are captured
exploited
utilized i'm not sure what word to use
here is that a
night like a reaction chamber are we
reacting observations and data
into inference in these reaction
chambers of mind
similar to the differentials that we
exploit
to in a heat engine for work are we able
to
to capture what are the devices of
capture sorry for the long question
right
yeah your hand on your chin makes me
think that maybe it's an intriguing
question
it's definitely intriguing i probably
can't do justice to it but um
i'll try um so just on the last point i
guess i think that the
um well actually so first to take your
your question about information
um whether we're using it um

so are you sort of like whether we're
really exploiting it
up to its maximum possibilities right as
information
or just or just treating it as sort of
you know
uh something like warmth a source of
warmth i mean i
i take it that insofar as we're using
information
so i right so it gets to the heart of my
perspective and to this
perspective of this paper i guess so i
said i flowed this thesis as a
possibility but i
i do kind of have some sort of
deep-seated conviction that that
information can always be looked at
sort of in terms of what it represents
and and physically right so any process
of information
uh information is literally a process of
physical change as well as
as well as a process of some system
coming to know something
right so to take the etymology right
just go back to shannon's original
framing
information is just the amount by which
someone or something has been informed
in my in my view so um so i think
insofar as we're treating something as
information
we are using it as information um
right it's not now i don't know what the
announce maybe i'd love to hear more
from you on this i don't know what the
analogy

would be to using information just as
heat is it like
we're aware of something but we're not
really doing anything with it like in
terms of inference
or uh so so i guess
my my view is that we are in so far as
we're thinking
and exploiting information sources for
inference we're using it to its
um maybe not maximally efficiently
always but we're using it
in a way that's that's intelligent but
but just one more point in this like
um the the analogy to combustion engine
and like
the the idea of um
i guess my thesis here is that anytime
we're performing an inference
any system is performing an inference
the physical
aspect of that is something like
right it is an actual physical reaction
so there is always something
that has precisely that nature behind
any
uh use of information um but i think
everything i just said sounds to me
completely trivial as i'm saying it so
um
that's its power yes perhaps
what did you mean we can unpack that
because
it's known that is um not just simply
wasted but where there's a heat
differential
there can be work and potentially when
there's an information

differential there can be an analogy to physical work but in the informational regime so what motivates that line of thinking

for you yeah it was well and again i'll send around the

slides from 2013 but but the idea was that

it is not by analogy it really is a capturing by the mind of an attribute of a system right you look at a block of iron it has

spin it has temperature it has color there's all sorts of dimensions that it has

that are unexploited um and then we can look at them and exploit them for different you know magnetism with spin and then temper or electronic electrons flow through iron same block iron it's just sitting there

but you measure different attributes and as you measure different attributes you measure them into existence and that existence then can be exploited

and so we did that with heat right they thought that heat was caloric they actually thought it was a fluid

right and that it but then the the guy who is the cannon borer i forget which one the guy who board the brass cannons

he said wait a second it can't be a fluid because it's unlimited it's a quality of the material

so i think what we're talking about in information it's a quality not of the brain

of the it's quality of the mind and the

differential between the brain
and the individual and the mind there's
a there's a
mind out there to be learned let's call
it an environment if it's animals but if
it's humans it also involves language
and when you're born you don't know it
and then you go learn it
so your brain has to get tuned into the
mind so to me the mind is our combustion
chamber
and you gotta learn the combustion now
if you do if you're different learn
different things if i'm a hunter
gatherer i'm going to learn different
things in my environment to combust
right i'm going to have different cues
and i'm going to say wow this is
relevant for me because i got to do this
and this
so you the we train i think we train
our combustion chamber it's symbolic
right since we're training this thing in
which to contain
the differential accomplishment
of the active inference the beijing
accomplishment
and then we trained so it was an article
in just the last thing i'll shut up
for a minute there was an article in
nature magazine a couple years ago they
said less than 12 month old human
infants learn in a beijing way if you
point to it if you tell a
baby a horse has four legs the
baby will point to a table a coffee
table and say horsey
then you say no

scott just disappeared it was too much
of a truth drop with the table
and the legs so stephen to you and we'll
return to scott when we can
okay please sorry okay scott's back
anyway continue with
okay continue from the table okay
yeah anyways bayesian you they beijing
accumulation you they'll point to a
table
then you point say you know let's have
hair they'll point to a dog and say
horsey then
they say no it has to have a main
they'll point to a horse and say garcia
so it's a beijing
the article was about bayesian assembly
of human knowledge
and so what if the baby is not really
assembling the knowledge but just tuning
their brain into the mind and the mind's
a combustion chamber
and the mind we teach we have to learn
the mind because that's the useful
combustion we've already
designed this community mind
to navigate space and we teach it to our
offspring in order to make them more fit
anyway else cool thank you
any thoughts on that alex well it's
interesting i
i don't know if i completely understand
the way scott's using the term mind here
but
i think i kind of can see how something
that deserves to be called the mind
would emerge
through this sort of learning process

right so so i guess there's a trivial
sense i think in which
anything that has energetic properties
has a mental side
that's kind of what i'm saying in the
paper um
but it's trivial and if you want
something that really functions
cognitively
then you need to have internalized a lot
of these regularities through
you know the sorts of learning and
infant schemes that we're
familiar with here so um if if that's
what you
mean by learning the mind i think scott
you now guys
analogized it to the environment at some
point or maybe it's more than an analogy
um it's like the the data the the source
right of the data sensory data that
you're interpreting um
so uh yeah i think it's a cool idea
cool stephen welcome blue
steven with a question and then anyone
else with a question or blue you can say
hello
yeah one question i'd be interested in
is um
do you see maybe more than one um
type of information processing that may
be happening at different levels
so maybe you've got because we've had
people talking say at the quantum
level which could be happening
particularly say the photons coming in
with the iris
and things going up and then you've got

other levels where you've got these
steady states
going at different rates and then you've
got
some of the things you're talking about
where maybe it goes into
this sense of a more averaged closed
mind idea which would have to be somehow
created maybe because
the dynamical nature of the data and the
noise is not necessarily
exactly how it is when it first arrives
at the organism
but maybe you're you're you're bringing
back some of that more
averaged out heat ideas
um around how energy can be averaged
over time
in in some sort of heat form is that
would that be accurate
um i'm not sure about the heat piece i'm
just
genuinely not sure we can talk more
about it but like
as far as multiple types of information
processing
i mean i think that's definitely what's
happening in the
you know in the brain of course and
there's all this
literature on on multi-scale active
inference and stuff which
my paper isn't directly about active
inference but um
uh i think like there's no so we tend to
signal out certain processes in the
brain as like
processing information but really that's

going on

on you know uh in on fine grade levels

uh and on larger scales as you say um

i think we tend to just pick out one

piece to analyze at a time i guess

the part of the question i don't know

how to answer is like what

what are the types of information

processing in question so

um i guess i'm skeptical of that type

distinction not not against it i'm just

not sure

what what kind of types we have in mind

but um

definitely i think there's a there's a

rule for averaging uh just in

if we even if we just stick to neurons

and don't worry about

underlying chemistry or physics like uh

if

it can't just be that one neuron

inspiring uh represents

you know your grandmother because then

if that neuron's shot then you can't

represent your grandmother

so um i think there has to be a

if we want to associate content with

brain states for example

those states are going to have to be

couched at the level of averages in some

in some sense or some kind of

statistical uh property

um and i i think that comes out clearly

also in the work that carl firsten and

uh banyan uh and others have done on

on the whole markovian modism idea so

they talk about the fact that it's

um uh you know it's um it's not

individual brain states that encode
information on their view it's it's the
the mean of the distribution
uh conditional distribution anyway going
a bit into the weeds on this but i
i i do think that um that's important uh
heat i don't know let's talk more about
it
yeah keith we had the what is
temperature slide which i'll
just switch to yeah yeah wait stephen do
you have any other thoughts
well you know don't slide and maybe just
as
just a sort of following from that is
maybe
something about the mind and the level
of the psychophysical
maybe actually the mind is kind of a
psychophysical level
as much we think of as a brain level but
maybe what you're pointing to is
something a bit more of an of a dynamic
um which is kind of interesting yeah
i mean i i generally sorry i generally
have trouble seeing distinctions between
levels so
so maybe that's what you know you you
all could help point out
like why these things are not the same
uh and it wouldn't would help me
go ahead hence the psychophysical
identity thesis
paper that you wrote right because
that's the entire topic of this
discussion
is what are those relationships and are
they

like each other or are they
overlapping are they isomorphic are they
one of the same so
um here on this temperature slide
you wrote about how increases and
decreases
in temperature change the variance of
distributions
so maybe just what is temperature or
in this psychophysical identity mapping
what does temperature reflect because
changing the temperature on the brain
you're not just talking about going into
a warmer
room you're talking about something
related to inference so what is
happening there
well right i mean i guess i am saying
like
look if if you change the temperature in
the room and that ends up changing the
temperature of the brain
then it's kind of an implication of the
thesis that
you are changing the temperature of your
statistical model as well to some degree
right probably usually to an
imperceptible or
insignificant degree when something like
walking into a different room
because now the brain has its own
internal homeostasis and it's not like
we're gonna drastically directly affect
the brain's temp
internal temperature um by changing the
room temperature one degree but i really
am uh
you know speculating or i'm considering

the consequences of
i mean the overall theme here and
temperature is a special case but the
overall theme is that
uh anything that you want like so any of
the terms in the home holds free energy
from from physics from thermodynamics
um should directly this is a transparent
encoding idea it should directly sort of
be reflected in the cognitive dynamics
and so the cognitively speaking what
temperature means
like it says here is just um uh
you know temperature parameters in
models neural network models and other
sorts of models um
i think i've heard this referred to in
the active inference literature as the
as the shaky hand parameter
right so it's like you're reaching for
something if you increase the
temperature parameter there's less
certainty
and exactly where the motor system will
end up right so
that's that's what it means cognitively
speaking uh
or if you don't want to call motor
control low level motor control
cognitive then
in terms of of these models of organisms
what it means is just
literally just the variability of the
distribution
right the variance so um i guess what
i'm saying is
yes i think if this thesis is true then
what encodes that

is the actual temperature of the system
that implements the model very
interesting thank you we'll go to blue
and then
scott

good morning i'm blue knight i'm a
research consultant
based out of new mexico and i really
enjoyed this paper i really enjoyed
the possibility of having a way to
separate
the mind from simulations of the mind i
think that that's a really
um relevant and pertinent idea to put
forward

but so in in relevance to temperature
and this is going to like totally throw
what i just said out the window
uh you know i i um it dabbled a little
bit in retail investing right and so
like the day before or like maybe in the
morning

you might like what's called take the
temperature of the market
right so you see kind of as things going
up or things going down like you just
kind of put your feelers out there
and take the temperature of what the
market's going to do

so i just wonder in terms of cognition
in terms of active inference if there's
a way to

like take the temperature in in maybe
like more of a

um of a metaphorical sense

right does that make sense

um i'm not sure um

so just i can elaborate a little bit

more like what uh
like in in the political system right
like
we get like a temperature like is this
like a liberal environment or is it like
a more conservative environment and that
might influence
our actions right like the active
inference loop might be
influenced by the perception of
temperature
in the environment right
so yeah i mean uh first of all hi thanks
for joining
um you you and dean and daniel uh you
all
i really enjoyed your discussion but in
point zero and
it made me excited to talk about it
today because it's it's cool how
just how many different questions or you
know anyway whatever it's cool
so um i mean sure you could use
temperature in a metaphorical way as
well like
um i think uh
it like i i guess in this paper i tend
to
i i like i tend to assume that
when people that that metaphors are are
not metaphors but identities right as
they graph
the ground level assumption and then see
see where that you're forced to
distinguish between them
so like if we're talking about again if
you're talking about a model of motor
control where you're literally reaching

for something

then i think in that case it's plausible

that there's some relationship between

the actual temperature of the pieces of

the system that are driving the motion

and the temperature parameter in the

model but like if you're talking about

something like

taking the temperature of the market or

you know the political

there still might be partial mappings

there

right like but you know if you're

defining some arbitrary dimensions so if

the the temperature politically means

one side of the scale is is conservative

and the other side is liberal

then then i wouldn't expect that there

would be some physical component to that

um but the same way that your paper was

uh as you said just putting it out there

not endorsing the psychophysical

identity it's almost like

an econophysical identity thesis

would be asking whether the thermo-like

dynamics of markets actually have some

sort of

mapping that's more than a metaphor but

it's something that can be stated and

blanketed or boxed off and then can be

understood as being adequate in certain

situations but maybe inadequate

in other situations but when we talk

about the market heating up

people know what that means in a way

where you know the market feeling happy

doesn't really capture it heating up or

cooling down so

scott and then anyone else with a raised hands

oops i'm not sure if i should try to share my screen i may just screw up so let's not do that but let me so i just pulled up my slides from 2013 and so

i said under the first law of thermodynamics temperature has two roles first from the perspective of a party

outside the system temperature tells whether two or more systems are in thermal equilibrium

same temperature equals equilibrium okay from the perspective of a molecule in the system

temperature tells about the distribution of various molecules in the system over the possible energy states right the higher temperature equals a broader distribution of molecules in different states and a higher boltzmann distribution so what i'm

asserting in this context in the context of those earlier slides was analogously here

are we looking within this system of the cognitive system and the dynamics of components within it

or vis-a-vis an outside voice

for that system i think that's a relevant question here because when you're talking about

the first law in the in here

we don't we have um this is um

um um maxwell's demon is actually a person now

sorry that obscure reference but
maxwell's demon was a convention
to plus with the second law of
thermodynamics a gating function
um and now we actually have
the notion of what does a molecule in
the system think
in a sense is what i'm inviting here and
this and so i think the assertion and
the assertion certainly i'm making that
context is we can capture
either one of those dynamics and it's
info dynamics you're not thermodynamics
is what i call it
but essentially first law tells you
about
interoperability is what my assertion is
because
you can have equilibrium in a symbolic
way i
assert is the equivalence of
interoperability symbolic
and so that's when you can have a
beating of the minds you can have shared
consciousness society
shared language all that kind of stuff
is my notion of where the
first law takes you thanks
that's really intriguing um i need to
read
your work on this please um
yeah i mean not to give a too hand-wavy
an answer but i think
since we're dealing with multiple scales
here we could talk about neurons
performing active inference
people doing it i think both of those
sort of uses of temperature are relevant

um

but exactly yeah um

we'll get to stephen but maybe just a
general question where does active
inference

come into play and how does it come into
play

oh yeah so um right so you

you guys raised in the point zero that i
didn't really talk about activations
that much in fact i only referenced the
literature on active entrance

um so this is yeah it's interesting this
paper isn't really about active
inference it's not

really directly about the fep either so
thank you for having me despite that
um but it's i mean obviously it's in the
it's in the very core it's concerned
with the same core issues right

but um i don't want to take us
on this if it's a tangent but the reason
that i don't write about active infants
that much is that i think from a
high enough or meaning abstract enough
point of view

um you you don't need i don't know that
there's really a difference

to me active influence kind of relates
to the stuff the way that like dawkins's
extended phenotype

proposal relates to more traditional
ways of thinking about

um evolution and genetics in that it's a
new paradigm

but i don't i don't think that there's a
system that you can describe using
active inference

that you couldn't also describe um using
say traditional
representationalist terminology so
that's obviously controversial
we could talk about it but um like
in so far as your as you're concerned
with analyzing or designing a system
like right like if you wanna even if you
want to write an active inference agent
as a model you only have control over
its internal states
you can't directly control it needs yes
it optimizes
um its situation by acting and
intervening on the environment
but you can't directly control
how the intervention on the environment
plays out you can only control the
expectations for sensory data given
that you've acted in such a way and so
um
that's why i tend to just default to you
know it's what
jerry fodor called methodological
solipsism right which is a
very different perspective from what
people i think i'd love to talk about
this maybe
maybe they're more similar than i than i
think too but um
anyway so so uh like active inference is
great i think in fact it leads to new
ideas for technology and new models just
because it's a different way of thinking
about things
but when you're doing like philosophical
analysis like this
i don't find that it's for me often

necessary

to think in those terms but i would love
for someone to point out
something that i'm missing by not doing
that

well what we love here in the active lab
is hearing everyone's perspective
on active inference and also adjacencies
and synonyms and other languages and
other ways of expressing it so thanks
for

sharing it the way you did um steven if
you

wanted to okay yeah

so so building on that you you mentioned
the helmholtz free energy
rather than the gibbs free energy so you
sort of the broader sort of
energy that's available um

the the i suppose a question i've got
here is in

in this in that um actually go back to
that diagram just one sec with the
distributions there you've got just
you you've got both the difference in
the sort of the size the distribution
for the different size molecules
but you've got the average speed is
higher for

um hydrogen but yeah hydrogen or
oxygen it's it's faster it's the speed
is faster for

something that's lighter and that may
have an impact

at the local level when the actual atoms
are

hitting each other at the same
temperature but

when you take the average it's all
around the same energy
if that makes sense so i just wondered
if there's any
particular sort of relationship
to the way that the speeds
are happening and the way that might
change something at a certain scale
and just helm holds and gives free
energy
yeah um so to be honest i mean the
reason i wrote
i write about helmholtz free energy is
um
just because it's the thing that jeff
hinton and others
um used in their um
when setting up things like the
helmholtz machine um the gibbs free
energy
uh i'm just less familiar with it's it's
it adds complications that i thought
weren't relevant but if they are
relevant
then let's talk about them um
yeah could we maybe hear a little bit
from you just what is free energy
in the senses that you used it or how
would you describe it to somebody who
might be not very familiar with
thermodynamics
right um someone such as me for example
so
what i would say is well i mean i'm a
philosopher but um
uh so free energy in a physical and
thermodynamic sense as i understand it
again welcome i welcome any

uh kings in the other direction um
but as i understand it it's uh just the
energy that's available
for doing work in this
the system so it's the internal energy
the all the energy that the system
harbors just in virtue of its
configuration
it's sort of existence at that moment um
minus all all of the energy that's uh
not usable because uh it's
it's tied up with with sort of this
disorder
um with with entropy um
so i'm sure that someone else here could
maybe articulate that
more gracefully than i just did but
that's that's the one term
and the other term in the in the
equation here is the variational free
energy
which is just um the difference between
a
uh the the basically the
the gen the gener the generative model
or the the joint distribution that
defines
the system's generative model of um
some kind of statistical model right
more generally and
um the approximation to it
that you've arrived at at a particular
time step right so if you're doing
variational inference you're trying to
fit this q distribution
uh which i mean you know people this has
been talked about ad nauseum so it's
hard for me to get the energy to really

launch into it because i feel like we
all
are familiar with this concept but it's
it's just the difference between the
approximating
distribution the recognition
distribution or density and the
degenerative density so so it's so i
guess the thing to underline there is
that it's
it's only defined in terms of these p's
and q's right q recognition
p generative that are they're defined in
terms of the generative model which is a
model right it's a
it's a it's a um a mapping between
like hypotheses right or hidden states
and
outcomes so there's obviously no a
priori connection between that and like
thermodynamics um
so it's just important to keep in mind
that one of these is is
in terms of how it's defined as
thoroughly like cognitive has to do with
belief updating and the other
has to do with the physical states of
the system and that mapping
between the cognitive and the
inferential and the physical
is again the entire discussion
and the entire point of putting out an
artifact that
is critiquable or is extendable in the
ways that you made it
so scott then stephen and then also
anyone else
who's watching live can just put a live

chat in and we'll ask the question so
scott
and then stephen so this is really what
a great conversation thank you we've got
a great paper
so couple things first first of all you
know i wonder we always talk about it
being by
analogy and it's not really
thermodynamics but von neumann
said to shannon use the second law of
thermodynamics for your quantitative
theory of information
because in his quote right his quote was
because the math is identical
which tells you something it at least
tells you something about math
and humans but may not tell you anything
about the environment but it tells you
something about math and humans
but he also said and if you use the word
entropy in any argument you'll win every
time because no one will know what the
hell you're talking about
so that's good advice on both of those
points right
so i always look at it and i go okay so
let's talk about those differentials and
you know we took that free energy
question that we just answered i
so i deal in policy i'm not a biologist
i'm a lawyer
okay in a physics lab and the reason i'm
interested in this stuff
is i think you can access the external
states
and in fact i think that's all we do as
social beings

is we that's what rhetoric is i am
constantly trying to
lasso external states of everyone else
that's what politics is that's
all the whole misinformation problem
right now we're setting out
now in a biological example we kind of
don't send around memos as much
now there is there are communications
obviously in
in the sciences and in the nature and
all that but when we get into the
human populations
we may have something more going on here
maybe mind is an emergent phenomenon
merely of the attempts at harmonically
coupling with other minds
maybe a break if we're successful
in tuning into that local mind we
survive if i
am dropped into france and i don't speak
french then am i going to survive
now now i will but what if it was 300
years ago or whatever right
so you need to if it's if the mind is
fair
feral and is trained maybe
that ability to affect the externality
is everything
maybe that's how we are humans in the
way that we are because we're pretty
pathetic little organisms on our own
each one of us right we don't have sharp
fingernails and our tooth and claw is
not
would get bloody but it would take a lot
of biting because they're kind of blunt
right so anyway that's a bit of a

tangent but
so the question is are we maybe if we
look at the
maybe we're not entering another brain
maybe the mind is where the action
is and when we're doing this in training
maybe we're merely tuning into something
that exists already
in our in our attempts here at this for
active inference and we're
just getting on board with the existing
mind and so therefore our steady state
our homeostatic state is the one that is
most aligned with the externality
that's drawing us there and as lawyers
and as policy people
we create artificial externalities
that's why i use synthetic intelligence
in our title
we're creating synthetic externalities
and saying hey come on over here we've
got a good externality over here check
it out
this one's resilient sustainable and the
sdgs in the u.n love it
right so we're creating realities what
do you think about that in terms of the
notion of the uh efficacy
of the possibility of harmonic coupling
with other minds and that as
a another active activity in active
energy is not just changing your
internal model
yes it's extremely interesting so um
i mean i think this question of how
exactly the external states get into the
picture is
is like pretty fundamental for a lot of

people who are in this area
and uh i i think you're right that it's
it's it's essential but it's essential
in a strange way where it's completely
inessential too so like
if it's it's essential in that the yeah
the reason that your steady state is
what it is
essentially involves the environment
right so like in the free energy
principle the
the next the non-equilibrium steady
state is defined with respect to the
situ
this you know a sort of a subsystem
that's embedded in an environment and
that steady state wouldn't exist without
both pieces of that
right and it's the fact that you're
doing gradient descent on that
that guarantees that your variational
inferences is going
anywhere useful uh and and that's that's
sort of the
sort of you know magical seeming
connection between actual
accuracy and and all this just churning
that happens when you're minimizing free
energy over time
um but but the the way in which it's
inessential is
of course i mean i i just i guess maybe
i'm
maybe i've been called a cartesian
before maybe i am i don't know i didn't
mean to be but
um there's a sense in which simply it
just the datum here is just

that i can get things wrong and i don't
see that there's any obvious way of
limiting the scope of that
right so just fallablism um so
i think there has to be some uh subset
of states of the overall
universe that you'd identify with you
and
your perspective and otherwise right you
have
sure the external states are essential
to this story
and they drive the dynamics de facto
but there has to be a difference between
the actual external state at any moment
and the external state that i represent
by my hidden
states or my my internal states um and
so
i think you know
like the reason any of this works in
prac in practice
is because we've internalized the
structure
of the external environment and this
whole point about harmonic
coupling between minds is really cool um
it's there's something really cool and
bootstrappy about that it's like
right minds are things that harmonize
with other minds right and
um i don't just to kind of
pick out another piece of what you said
i um
the way that i tend to think of this
stuff is in terms of sort of imitation
so the fundamental term for a lot of
people in this area is prediction

but i almost think it's more like we
just learned to our brains learn to
imitate
the the sensory uh right the patterns of
the processes that produce patterns of
sensory input but the reason it's not
just imitation is that
also the world is just made up of a
bunch of these generative models that
each have their own driving right um
signals so um so that
i'm not sure i don't know i don't know
how to tie that up in a nice package but
it's it's a cool set of questions
um and uh
but so so the reason though that i
stopped short of saying that the
external
states are as fundamental
in understanding cis systems biological
systems uh
people etc as the internal states is
simply
things like cartesian skepticism so we
could talk about that
um well yeah go ahead scott just a
follow-up
well it is interesting because it's um
because our struggles in it
when we you know i say on this this
group very often
the only time you see reality is when
you see paradox
right if you're seeing absence of
paradox you're intrinsically seeing a
model of some sort
and so we should embrace a confused
state and and know that that is a

reality

and so that is not just a statement for

this conversation but in general

maybe what the mind is doing is the same

thing that the body does when it's

standing still you're actually never

still

you're always dynamically moving to

maintain

a semblance of stasis in a banded

stillness right

so maybe we have a banded mind maybe

it's fully dynamic

and maybe what we think of as us an

existential kind of element of it

is within a band right do people change

over time it's a ship theseus in a way

right you're still the same person but

you're

some of the stuff has changed out

swapped out over time um

so maybe the embodiment helps us with

mind as a fractal way at any

level because we can say okay if we have

a system

that's doing processing of dynamic

environment some of that processing may

be internal some may be external

i mean that's group formation and you

know

part of its genetic and consanguinous in

terms of performance and some of it is

people who are similarly situated right

if you're in a disaster and everyone's

all of a sudden has the same problems

people band together

all of a sudden very easily and their

mind morphs into that

organism that is needed for that acute situation

in that stressful environment and so they maybe it's always dynamic but that we don't

see the dynamics as much and they seem more leisurely when we're studying them in the abstract but maybe they're highly dynamic in both directions especially

as we become information beings and more dependent on symbols

thanks scott just one thought on the dynamics

is anyone who's recorded a neuron or thought about that knows that just to keep an idea in your mind

for a few seconds is being underwritten by

vast numbers of dynamical processes that are quite rapid

and the second point about the internalities and the externalities and the internal states and external states the helmholtz free energy has this internal energy

and that's like the configuration of the molecule

usually thought of as and then this negative ts

is sort of that those vibratory modes that can't be as

easily harnessed but when you pointed to synthetic externalities

which is the same thing as saying reconstructing the internalities

it's like what if all of a sudden the organization of the system

we could partition it a little bit differently so that we had more u and a little bit less ts and when you think about a vibrating molecule u makes it look like the covalent bonds and ts is the vibration but if you think about an ant colony the movement of the pieces is still part of the internal dynamics and so especially when we're talking about information which isn't covalently bonded atoms internal energy is something about the model it's not just about how things are touching and can be broken and reformed so there's a lot of awesome pieces in what you added there steven and then anyone else with a raised hand yeah i think building on that you're right the there's this in this i think you're making the right choice here for this paper using the helmholtz free energy i think that's a um even if gibbs free energies use more in active inference that adds an extra level of complication and i think this is actually useful then because you're able to look at some of these distribution questions and just how information is extracted but it's actually helpful for me to see you go back to the helmholtz because it makes me realize well that gives free energy which is more

about how action happens how do you know
what the internal energy change would
have to do
if it was trying to also keep within a
certain constraint of volume
and pressure which you don't have to
worry about
with helmholtz free energy that would
bring in action that would bring in the
action part of active influence
in some ways um but the underlying
principles
still hold for what you've got you know
so i think that's um
that's helpful so i don't know if you've
got any thoughts on that yeah
yeah that is health i would just i would
just say that that is helpful and um
uh i would love if in the future
anyone
yeah worked on this same topic and one
and generally you know
ex unpacked the the sort of um
i was going to say generalization but
it's maybe the opposite um the the
how this would work out in the case of
um
of action more specifically and using
gibbs free energy and stuff that would
be really cool
awesome and taking the temperature of
a person in terms of going for a checkup
to the doctor
and they put the thermometer under your
tongue and that's a thermal temperature
and how are you feeling what does it
look like to take the temperature just
like

blue had been suggesting so maybe let's go to the transparent code um slide that you added alex because that was definitely something that we were curious about from the dot zero and we can just hear your thoughts on this talk a little bit about this and then also anyone else can just write down a question raise their hand or ask in the live chat but we'd be really curious to hear what you thought about this

yeah sure so um yeah so the reason that again that i put this in was that dean and um and maybe maybe everybody in that point zero um discussion had um had questions about this and i think uh the transparent code it ends up sounding like a slogan that invites you know speculation and it's like it's a mystery or something but um this might be a case where i'm trying to say something so simple that it's like

it's easy to miss right so so all um the role let me just first frame this the role that this plays in the paper is that if you want it to be that variational free energy and hellhole's free energy in the thermodynamic sense um are the same quantity then you're going to have to um explain how well both of them are sort of made up of probabilities right so you need to

have a mapping from one set of probabilities to the other
um so uh there has to be a way of understanding how the probabilities of states this is a big distinction in the literature right how the probabilities of states could represent could represent so the probabilities of states in the system the cognitive system that you're analyzing which is also a physical system
how is it that those could represent um the probabilities that are assigned by that system to the external states right um so so the idea of a transparent uh neuronal code or any code uh is is just that the probability of for example so in the on the left hand of the diagram here if we had a transparent code in the sense in which i'm using that term then for example the probability of neuron b firing given that neuron a fire is right so if b is postsynaptic neuron and a sends a signal to b or a a fires and uh there's a certain probability of b firing subsequent to a firing that would represent uh the probability of some state of affairs q obtaining given that some state of affairs p obtains where a and this we can we can talk about where this mapping comes from right but

the assumption is a
represents p occurring and b represents
 q occurring
um so so that's really all i meant by a
transparent code and it's it's not
it's a simple idea but it's it's
definitely controversial right like
that that would mean that if you think
that there's a certain conditional
probability of
you know the cat being on the mat given
that the cat
lives in the apartment um then
sort of the probability that you would
represent those things has to be tied up
with the probability that you assign
uh to those things occurring and that's
not at all a
sort of um uncontroversial statement
um but it's a simple idea and this is an
idea
by the way again that's that's used in
some of the early connectionist
literature
um on um blue mentioned um
boltzmann machines right so there's an
early paper on both both of the machines
uh where this kind of stochastic code is
proposed
and the basic one more sentence on this
the basic idea is that the objective
probability
sort of measurable probability of events
within the system
uh maps onto the subjective probability
that the system assigns to
the states of affairs that are
represented by those internal states

interesting and in the context of the
cognitive
system of interest we're talking about
the brain
it's like we know that we're not going
to get the transparent coding
from the photons hitting the retina to
what is the probability of
the cat being on the mat so with this
multi-level
processing as you've kind of hinted at
with the multiple layers here
it could be possible that at some
neuronal stage or
at some level of processing there is
a pattern that maybe represents
the cat being on the mat
probabilistically which also invoked the
whole representation
questions which we had talked about
months ago
um right so it's so fun to see how you
you delineated where there's controversy
sometimes even both sides of the
discussion around a controversy
and then it's like great here's the
artifact let's see which way we're gonna
push it
if there turns out to be more evidence
for transparent coding
then we'll push it that way and if there
turns out to be less then we can push it
a different way
so um scott uh if you're
there i don't see you yet so stephen
first and then
okay stop scott go ahead and then
stephen um

i just wanted to mention there was an article in nature a couple of years ago where there was an assertion made that consciousness this is what caught my eye consciousness they asserted arose 520 million years ago and they and i was like well that's interesting what's the evidence and what they found was some structures for i think they were proto-nodo cords or something but basically it was a ganglia kind of thing a secondary processing and their assertion was that because they found physical evidence of a neuronal structure again in in fossils it was some eddie karen are able to be preserved in shale or something but um uh that because it was secondary processing beyond the the eyeball or the ear or the perception apparatus that secondary processing they suggested meant there was consciousness because that was their definition but it was interesting the reason i raised it here is it's interesting that notion of embodiment right so we're talking again let's talk about reaction chambers we have there's stimuli all over the world there's things happening and then we have perceptions ability to detect stimuli right in our afford it to us and so that comes in but we don't have the

structure of it is
tbd right and so then we the the
the organism embodies morphologically
certain structures over time eyeballs
and
notochords and ganglia and things like
that right
and those themselves are information
structures they just haven't even
embodied in a morphology so it's
morphological computing for biological
organism right because they now can do
that thing i can't see an ultraviolet
but bees can that kind of thing or
infrared that kind of thing um so it's
so i just wanted to raise that idea
that assertion that offended me
originally that consciousness arose 520
million years ago
is not so offensive in a way because
what they're trying to do is grope at
the notion
of the layering on of morphology and
embodiments
and say look there here was a ratchet
here's something happened
here and we think we don't know how much
more experience it's like julian james
when he went back and did the
origin of consciousness and the
evolution of the bicameral mind book in
the 70s
j-a-y-n-e-s and he was asserting from a
literature search
that the wernicke's area flashed on
at one point it was equivalent to
broca's area language center in the
other side of the brain

and his literature search he looked at
joan of arc he looked at the iliad
and he said that morphologically
literature
embodied that notion of the mind
actually changing
ancient literature can see the mind
turning on that internal voice and so i
just wanted to raise that
notion those embodiments of the external
and internal states
lying around all over the place right
and so
it's your paper does a nice job of
inviting
the um treatment of those or those
those as candidates for looking at the
mind being assembled particularly
if the mind is external to the brain and
the brain is just an antenna tuned into
the mind
thanks one one funny little
synchronicity there is that
uh julian janes wrote the book that
scott mentioned
origin of consciousness in the breakdown
of the bicameral mind
but edwin janes was a very famous
thermodynamics
researcher who introduced the maximum
entropy
interpretation of thermodynamics so
related like we all are bi were they
related
were they related by cons the blood
aren't we all
yeah oh come
yeah just sort of coming to this diagram

that you've got here um
so i think um one thing i'd be
interested in
unless you say neuronal or other
internal systemic states so you sort of
give
a little bit of latitude to dynamical
aspects of that
goes to cognitive states of the system
beliefs
and i'm wondering how much
you're tied to those beliefs being
um statistical computational
sort of separate things or how much is
it
beliefs on action policies i mean i know
i'm pushing active information again but
yeah it's you know it's fun
yeah what's your thoughts on that yeah i
mean
there's all sorts of questions here
about how we individuate representations
and such but
i wouldn't think i don't even i don't
really think of these as detachable
uh separate entities at all so that node
is labeled p
but i'm one of these functionalists who
thinks that the reason that
the label itself does nothing right the
reason the the label is a useful
indicator for people outside the system
to understand
what it represents but the
representational function is just the
role that that node plays in the network
so um um and that would i think that it
subsumes and includes um

connections to action right i just tend to think of connections to action as the special case where what you're generating what your generative model is generating are states of the motor effectors

so right i just and i think any any kind of planning

um that that that the organism learns to do

can also be understood as learning sort of sensory motor contingencies which will

influence how states of the motor effectors are

generated in tandem with states of the perceptual system

right how perceptual biases occur so um

basically yeah i don't think that

that there's any like these don't need to be cognitive separable

features of the these don't need to be separable features of the system they're just

like when i say that a state represents

p

um because i'm represents for me has sort of a functionalist meaning

i just mean that's just sort of a label

for uh

a set of uh ways in which the system can behave

because of its internal structure so

including sort of internal

um dynamics right but it's really just a

label for

a property of the dynamics of the system

cool

and also beyond the action being at the gross organismal motor level potentially internal representations or states could be like hormonal or neurosecretory so the neuron can fire but neurons have other affordances like the little micro motor behavior with the motor proteins like releasing a vesicle of of a protein that's been transcribed and translated also tiny little motor behaviors of nested smaller agents inside of that neuronal interface so stephen and then anyone else with a raised hand actually just what you were saying there daniel does that mean that you can almost release hormones through the brain it doesn't have to be like through a gland in the body but you've got smaller scale releases neurosecretory cells and neuroendocrine biology are large areas so yes many kinds of cells pull in and secrete molecules and neurons are of various subtypes secreting specific hormones neuropeptides all kinds of hormones and signaling molecules short range and into the circulation okay thanks um and i let that process and that's really useful i didn't know about that

and um i supposed just adding on to this
you you when you mentioned about sensory
motor contingencies and
and and being able to um so i'm
wondering that the scale
of um consciousness
you know and things entering into
consciousness or the phenomenological
sort of awareness
do you sense that there may be even if
active inference is under writing it all
some other types of like it doesn't have
to be as pure active inference at that
scale
and it could be working with other types
of
sort of architectures or process
architectures
uh i mean well one thing is that
consciousness
so i think scott brought up
consciousness for the first time here
that's really interesting i um i sort of
deliberately didn't say anything about
it in my paper because i find that it's
um it's a very thorny topic for me of
course for many people right
it's not it's not notoriously not an
easy problem or maybe it is and it looks
like a hard problem but anyway
um yeah so so i'm not sure i i do think
that like
all of the cognitive dynamics i'm
talking about are probably more
fine-grained
than what shows up in consciousness so i
wouldn't say that there's some
transparent mapping from

whatever you believe in in the sense of
belief that's implicitly defined
here and what you are consciously aware
of
um i so we can there's all sorts of
theories of how consciousness
relates to this stuff um
there was a second piece to what you
said that i wanted to address but i
don't remember
what it was well i suppose there's a
scale but i think you kind of addressed
it a little bit there
sorry daniel you're gonna say yeah we'll
we'll come back to it if you
remember what it was yeah scott and then
anyone else with a raised hand
yeah i wanted to go back to the notion
of neurotransmitters from a minute ago
when you're talking about endocrine
neural function
a couple of things there's a couple of
books here just want to hold up this
cognitive
um cognitive archaeological
the cognitive archaeology and human
evolution kind of an interesting notion
there's a couple authors there
here's another one archaeology and
memory kind of an interesting one
the reason i raise these is kind of
interesting when you talk about
neurotransmitters i remember reading an
article a few years ago
that suggested that among all the
animals in the animal kingdom now the
animal kingdom is an old
representation corollas linnaeus thing i

get it but in any event
um there's only one animal that doesn't
share any of the neurotransmitters or
the rest of the animals like the box
jellyfish or some kind of jellyfish i
forget which one it was
and i thought that was interesting just
what we were just talking about a minute
ago daniel was noting about
neurotransmitters
well again it's morphological computing
those are artifacts
of earlier solutions their practices
of processing of information by these
organisms right
and so the fact that we can go back and
say
wow i wonder if these things were
related given the fact that these don't
share any neurotransmitters that's kind
of an ethereal element it's not
something that you get like bones and
you know it doesn't fossilize well
but isn't it interesting that we can
find the morphology of neurotransmission
in current organisms and use that to
understand
lineage and relationship so in fact
that's a it's a morphological artifact
and it's not epigenetic
right but it's interesting the
relationship of
the the uh homeostatic state of
uh thinking the cognition of a certain
line of organisms
and the morphology of its
neurotransmitters
i'm interested in you social insects for

instance

do people do profiles if anybody here
knows anything about you social insects
please type in

the um do they what's the story with
neurotransmitter

analysis among you social insects is
that a special

neurotransmitter specially invoked and
is the morphology of neurotransmitters
perhaps

where the organism resides more than
other physicalities

thanks great great question scott so
first on the

tina four point and then the eu social
so um the tina four indeed

it did suggest new evolutionary
hypotheses

when people investigated which
neurotransmitters tenoforms used
and they convergently utilized some of
the same simple

charged amino acid derivatives but they
use them in different ways for example
glutamate

is not used in muscle contraction in
vertebrates

we have acetylcholinergic receptors at
the muscle

but glutamate was used which supports
alex's

point about functional representation
there's nothing that says

glutamate means muscle contraction it's
a functional

context with a receptor and a muscle
cell and a neuron that generates the

contraction

and potentially for a later day maybe

when we discuss

active inference the paper but um we've

done multiple studies measuring

neurotransmitter

metabolism and gene expression in ants

and there's been a lot of work on

honeybees as well and just the one

point i'll make here since it is a large

area and people can look at the papers

if they want

is that the neurotransmitters that are

involved in the regulation of foraging

in ants and bees are inherited

from the neurotransmitters that regulate

foraging and solitary insects

like dopamine and octopamine in fruit

flies where it's been investigated in

the context of solitary foraging

but the inputs and the outputs of

foraging

for the eu social insects are different

so whereas in

drosophila it's like internal hunger

signals representations maybe some sort

of neural hormonal

encoding leads to foraging behavior

and then that results in the acquisition

of food that re-enters that bodily

system

but an ant forager is foraging

essentially food for the larvae a

different part of the colony organism

so the inputs like the stimuli that lead

to induced foraging behavior and the

outputs like where the food goes

are different but the foraging motor

behavior is conserved
and it's that rewiring of the motor
motif of foraging
via the same neurotransmitters
octopamine and dopamine and others
that actually enable the colony life
cycle so it's a really awesome area i
hope we can
talk more about how that relates to
active inference
so um stephen and then scott and then
anyone else who raises their hands
yeah i'd like to ask a question
about which sort of links to that it's a
it's a kind of a semi theory i've got
but
is it feasible with this that the brain
like an organism basically the cells
that become the brain
effectively they mutate and they they
transform into the unique form that the
the brain and the nervous system
basically like another organism growing
into the main
organism right so you've got two
organisms once they differentiate
and their brain is like the control and
the
nervous system is like the control
organization growing into
and with the other the fascia of the
other organism well in a human fascia
but obviously
i just wondered if that would be
plausible and that that would do
anything with this brain
organism boundary question that i know
you

are kind of connected to in terms of
wanting to maybe maintain more of that
potential

i don't know if that's something that
other people have talked about or
whether it's a little bit out there
and just um yeah biologically or
from the sort of cognitive philosophical
sense how that pans out
yeah yeah i love i love the like
biology and biochemistry and stuff and
all this

you know wonderful fine grain scientific
um

life sciences stuff that you guys are
bringing to the table on this um that's
that's not as much where i'm coming from
but i think it's really really cool

um but so so that steve the idea that
you just articulated sounds
sounds out there and plausible to me so
um i think that like

uh that's kind of what i would expect
is that this same story can be told on
multiple scales internal to the system
right and so the brain learning sort of
learning

growing uh i guess

i've just been conditioned to think of
all this as one type of process right so
i said learning but

morphological development and growing
into its into its

um mature shape is a form of learning
but so anyway um

that happening and interfacing with the
rest of the organism um

i mean i definitely don't see any reason

that we need to draw a like
joint joint in nature around the
physically
right the organism that we can perceive
there are going to be many similar
joints that we could
delineate within so um i think that's a
cool idea
i don't know much about the details of
that process so i can't like
check that validate it but it sounds
plausible to me
thanks alex blue and then scott
so just to kind of maybe um ask steven
to unpack a little bit what he said so
i think um in terms of like a
brain organism like separation or
duality like
do you maybe mean like mind organism
because as
like in my experience it's like the um
like the brain is like the control
center right like i don't know that
there's necessarily like a
division between the brain and the and
the organism but definitely like the
mind is kind of like this ethereal
concept so maybe could you just unpack
that a little bit
well i was actually thinking partly
thinking about the work of mike levin
um mike levine or mike levin and um
so in a way i my understanding is that
once a cell becomes a neuronal cell
if there's no going back like it's kind
of
like it so it's in some ways it's um
it's it's i was curious whether that

whole

structure be it from a small organism

is effectively like we basically have

two organisms in our body

because our cells have differentiated

into the body

organisms and the neuronal ones and

they've sort of

they they work together in this

symbiotic kind of

cybernetic loop so that that's the kind

of thinking

um about how it pans out i'm you know

so i just if i could just like speak to

that a little bit um

so the neurons can de-differentiate and

do um in like especially like in neural

cancer types

so we do see some some d differentiation

there it's not as committal as we've

thought before i mean this it's kind of

a budding field and the

the you know antique way of thinking is

that there is

it's a set lineage but there's there's

ways that you can convince them to do

otherwise

i think mike levin's thing is the

bioelectric

field right and so so he separates the

the bioelectricity

of the organism as as maybe

the overarching control center there but

i've not seen like a neural

um phys like a neuron body distinction

that's why i was just like

what sorry well one thought on that and

also to scott's point about the

emergence of a second level of the nervous system is to remember from developmental biology that ectoderm the outer layer of the developing embryo turns into neuroectoderm which is the tissue that gives rise to the nervous system and gastrulation in the neural tube so the nervous system developmentally neurons share an affinity and a developmental proximity to the external layer and so when you start nesting internal layers inside of that tube that's closed you start getting higher order nervous systems and then also the pituitary gland to sort of close this with a neuroendocrine discussion the pituitary gland is actually amalgamated from multiple developmental sources and so it's interesting that it has such a pivotal hormonal position integrating hormones and the brain's function and it itself is like a re-fusion from a tissue perspective of multiple branches of the ectoderm that had branched and then they reformulate a new organ which is the pituitary gland that serves a very coordinating role in organismal development so Scott and then anyone else with the

question

you know this is the best way i can

spend my day

with people who say the quote the

pituitary gland is amalgamated from

multiple ectoderm sources

there i'm in the right place man you

know when you you know

you know this is the place for me i can

tell you that

hey um a couple things let's go back to

this part in holland dynamic and cancer

as an organism stuff for a second this

is uh this is good stuff

so you know the cancer analysis where

they start to treat cancer

as an organism in an ecosystem you know

some of the framings to think of it that

way a recruitment of a kind of chaotic

kind of like having a drunk uncle stay

at your house is one way of looking at

cancer right it's a it's a chaotic use

of resources

um so that is kind of interesting right

because

so we're talking and going back to

steven's point about the mind being

different

so i'm thinking of symbiogenesis here

lynn margulis stuff and the chloroplast

in all plant cells and the mitochondria

and all animal cells were once

their artifacts of once pre-existing

bacteria right they migrated on board

with eukaryotes because it was a better

deal

right more leverage more de-risking

higher levels of engagement with the

outside right they change their internal
physicality
and morphology so they could think at
way higher levels they could
access way more complex levels of
engagement
than single cell organisms including the
engagements they could get through
sexual reproduction which introduced
additional symbolic novelty so they
could be more dynamically engaged with
environments right clonal reproduction
does not introduce a lot of novelty
so it's hard to keep up with dynamic
environments if you're formal
this is something like intelligent
design but it's not so
it's interesting steven in your notion
of the mind being
separate because in a way the mind is
formed
by the social environment you know i've
told the story before my sister
adopting a five-month-old chinese child
who is now 22
who is raised in central pennsylvania
speaks spanish and english does not
speak chinese does not like chinese food
does not like chinese music
and basically looks like a chinese
person but is not so
is that mind a chinese mind the problem
is we don't get have any control groups
on the mind formation right once you're
formed you are who you are
and we say okay that's it but i'd like
steven where you're going on that notion
maybe the mind is an independent

quantity and
again the brain is an antenna to the
mind so maybe the body
contains the antenna components again i
don't mean a real antenna but sort of
um and it contained it needs to be tuned
it's feral and so maybe what's happening
is the tuning actually creates
a mind maybe the mind is not resident in
the brain
in a child and maybe bayesian inferences
construct a mind
and to maybe direct it to alex before we
go to steven's question
dean you really focused on this sentence
it may be for example that biological
systems
run an algorithm closer to of divergence
than to simulated annealing so what does
that
mean or what what does that reflect
on can i indeed maybe
give a little extra info on that just
just a quick quick thing
i want i want to hear stephen and i want
to hear everybody else
but i have a question in the background
sorry i have to say this in the
background i've got people taking the
front of my house off
so i haven't been able to do much other
than try to control my add
but i have a big question for alex once
we get through the
the part that stephen and scott want to
have addressing it's
specifically to what you just raised so
thank you

great let's go to steven and then we'll
return to dean and the contrast of
divergence
yeah maybe this will bring this together
is i suppose what i'm just
proposing is is that there is some
structural biological differentiation
which is will have an impact
in some ways and how we think about
extended cognition
but if we i i'm very much of
the mind body
environment dynamical system kind of
idea
so i suppose i'm i'm talking about there
being a difference
biologically but probably at the macro
level i'm
i'm trying anyway to formulate
approaches which talk about a whole
system
and not anything else than that but i
recognize that
it may open up some other cans of worms
but
um i i know that how i wonder if that's
something that you're
does this does this annealing relate to
the mind-body environment dynamic in
playing out within that or is
is it a little bit more separate or
different to that
yeah i mean the the use i was making of
this
contrast between contrast to divergence
and simulated annealing here is
was just was kind of more specific
um i was just thinking of

so annealing this process of of
gradually lowering the temperature
and simulated annealing was a
optimization technique that mirrors that
process

and um so i mean i guess the answer to
your question stephen is i

i wasn't really thinking about system
environment relations here at all i was
just thinking of um

two different sort of internal processes
and how they might be mapped on to
um onto these these uh learning
algorithms and processes and physics but

so

so i mean i can wait to go on to that to
take dean's

question or yeah or whatever maybe

let's uh let's go for the divergence

yeah

i i'm not predictable

um i'm going to pull up the paper

because i need to uh

explain this so alex in the paper you
you

you talked in in the subsection one two
the role of internal energy

um let me just find it here

you talked about the inverse

relationship between probability and
energy

is exploded and most uses of energy as a
cost function

and optimization problems so you made
that clear

and then a little further down in the
paper you talked about contrastive
divergence which in equilibrium

distribution of a restricted boltzmann
machine
and then you got into a little bit more
about the the comparison is
the simulated annealing and then you got
into section two
and i'm just going to try to find it
here the ensemble piece
so here's my here's my question
um if i were to bring a um
if i already bring three things and and
cast the widest
largest wrist and blanket yet around
your paper
because i think that's what your paper
does if i were to bring a polaroid
picture a sprinkler and an astronaut
together so if i were to bring
identity which is what was the primary
attractor of your paper to me
identity astronaut that
that distribution piece and
um the first one the
the the polaroid the isomorphism
could you explain how you you brought
these things that are so
seemingly um distal
into some sort of a coherence how how
did when you were sitting down because i
know you did this on a very
proximal level but it it has far
reaching implications based on the
kinds of questions that stephen has
asked the ones that scott's asked the
points that
lou and daniel have made it it doesn't
just seem to be local
it seems to be really distributed as

well and so

i'm just really curious you wrote the
paper but now that you've kind of been
able to

step back from it and sort of take it
into perspective

that distance must have you really
really looking at this maybe maybe in uh
interesting ways because there isn't
just the stuff on the paper

there's the stuff now that you've kind
of reflected on and seen the inversion
and and wondered holy macro did i open
up a can of worms or

what did i what have i done i'm just i'm
curious about that on a personal level
right um well just i mean in general
my the most striking thing about how
this has played out

um is that people are really interested
um so

blue mentioned this in the discussion
and um

i had emails about it too are really
interested in the distinction between
um genuine minds and like simulations
um which was completely an afterthought
when i wrote this

it was just like oh yeah i better deal
with the scope of this and so i should
put a sentence about that and

i did put care into it you know i i put
the best sentence i could and but
it wasn't my focus but um

yeah the i mean the the

the the thing i guess the thing that
that strikes me about this so it was it
was

right i mean it was intended to be this
it was it was intended to be a very
fundamental like thesis that i'm
considering here so
i do think it has far reaching
implications i have no idea whether it's
true but
right if it did it would have at least
far-reaching implications i guess what
i'm
um what i'm less clear on is well i i
i'm still i'm still waiting for the
other shoe to drop i'm waiting for
someone to say here's an example of a
biological system in which
there's no way that you can map the
cognitive interpretation directly onto
the physical interpretation
because um if that's the case and if i
think if it's a compelling example then
then i guess i'm wrong so far so far
that hasn't happened there have been
some interesting suggestions
um but so so i guess i'm i don't know if
i'm directly answering
your question or dean i'm just
um to me the the the devil is still very
much in the details and
this is essentially a defense of a pop i
was trying to
i was trying to show that maybe some
some a priory
prima facie objections to this kind of
thesis were not
uh you know were not compelling
but there could be all sorts of details
that merge that
that might question it um

so i don't know if that really answers
your question i'm sorry no no it
and again it was kind of like when i was
talking with blue and
daniel in the point oh and i i brought
up so i i talked to people who
who very much are in an activist camp
and you know they want it they want to
see that
realistic and then there's the
statistical camp and they really want to
have that
sort of top down where's the channel
what's going on
and i i just kind of thought what your
paper did is it
it opened both of those perspectives
i don't know that they even have to
necessarily merge or overlap i think
the fact that we can we can
use both and see both it's just a really
interesting thing now
i don't know whether that is disprovable
or it should be
uh i guess that would mean that an
activism is not a thing or
statistical modeling is not a thing and
yet we use these things so i
i mean i guess they can be discounted
but i don't know that they'll go away
that's all i so i guess you have
answered my question we can leave things
open
and up and not necessarily have them
provide everything but but we can't
just sort of discount something and
throw it away
until somebody as you said comes along

and says no right
it doesn't work yeah and just one quick
follow-up on this this
the one reason i wrote this paper is
that i think
what a lot of people find compelling
about the free energy principle and the
literature around that is
the idea that there might be a link
between physics and
cognition right that there is a
fundamental theory kind of emerging
but it's i think it's very hard to pin
down what the precise relationship
it's hard for different people at
different times it's becoming
it's becoming clearer this has been
worked out a little bit in the free
energy principle literature
but like it's it's difficult to
articulate exactly what the
the deep connection is between
you know variational inference on the
one hand and and like
um the behavior of physical systems so
i just wanted to i you know i wanted to
throw out
here's one um version one possibility of
what
the uh supposed to be a clean precise
brief statement of what that
relationship
could be um so that you know it
yeah it's open-ended but so that it can
be you know false falsified juxtaposed
to other
um other interpretations of that
relation etc

thanks alex we'll go to blue and then if
anyone in the chat
or if dave has a comment then feel free
but first blue
so alex i think that this goes back to
what you were saying in the very
beginning about scale right so it's it's
it's hard right now even to map like
cognition onto like neural activation i
mean i mean right now we don't know
um too much like too many components in
the system
are are murky for it to really be
clear what's happening and i think you
know as we talk about
information going from a cell you know
to from a brain to a neuron to cognition
to action
to bodily regulation and these kinds of
things we don't really know
is information compressed is it
integrated like
what is happening to scale a system
up or down right i think that that's a
hard
problem yeah i mean maybe i'm not sure
what you mean by scaling the system
up or down um
in in this section you talk about the
scale-free nature
of free energy dynamics and about how
global and local free energy
minimization
could be understood as subsystems that
are
undergoing some sort of uh akin process
at their own skills
right um

yeah so i mean the the thing that
so i mean the thing that i wanted to
react to and what blue said was
um the idea that we don't really know
what's going on
in the brain still it's very murky i
mean i think that's true it's always
relative right to some
ideal of how much we could know but i
mean
what got me into this whole literature
in the first place was was i was
i started i starting again starting with
the machine learning
literature around 2012 2013 i started to
feel like we actually were getting a
grasp on
on some of these principles how this
works quite generally so like
like the helmholtz machine for example
um that blue you talked about that in
the point zero
um i think that's a model now if the
model is accurate
then it seems like what's happening is
as we
um you know as signals are sent from one
neuronal population to another
um further from the sensory motor
periphery
um the information is being compressed
right so that's a particular hypothesis
that
there's a successive compression of
information
meaning that things are represented
using uh you know lower dimensional
um well vector spaces if you want to

translate between vector spaces and
neuronal populations right
um and and that we have a pretty good
idea
right like this this sort of
architecture explains
so many things i mean that's this is
this is what got people so excited about
the you know
predictive brain or the bayesian brain
type hypotheses is that
there's so many different things from
different domains in psychology and
neuroscience that this seems to
this kind of perspective seems to unify
um
you know it's it's it's it's to be
expected in a sort of helmholtzian
architecture that like
you know uh the you can always sort of
endogenous
endogenously cause the same sorts of
states that you could be in
as a result of perception right and that
imagination
and internal generation of states has a
particular role to play in learning
and things like this so i'm kind of
rambling but um
my my sense is that this um
this sort of bayesian helmholtzian
revolution or whatever you want to call
it
in recent years has has shown that we we
do understand
many principles of how the brain
functions at the level of like at the
human scale

um if the question is about like how to
how how do other scales fold into this
i um i mean there's some work that
maxwell ramstead and casper has been
people like that have done
on on understanding their on all
population dynamics in terms of active
inference at that scale
and so that's really interesting um
uh so like what i would think of as a
structural like a structural
representation from the point of view of
me or you like a human being
um you can also think of as a process of
neurons performing active inference
right at their own scale um but i think
there's to me the hard question there is
how do you relate
uh the contents of the representation at
the higher scale to the contents of the
lower scale
that that's that's the thing that
bothers me about in this area
yeah just if i could just respond really
quick yeah that's exactly
the the thing i mean we can see how this
active inference is happening
at different scales for sure but how
scaling up and scaling down or
or the the translation from one scale to
another happens is very
like mysterious still yeah totally
cool so we have about 20 minutes left so
if anyone has a question in the live
chat
otherwise we'll go to scott and then
stephen and be preparing our thoughts
for what to continue discussing in point

two

so scott so um picking up in the last
couple of points

really interesting the i was thinking of
what was the gentleman's name who
barrettson who said information is the
difference that makes a difference what
was that guy's name is it

anyway the difference that makes a
difference and i always thought
the first difference that he's talking
about is the

very um perceptual a perceptual change
in the environment and the second
difference is

is that perceptual change relevant to
you and so i want to unpack that a
little bit here because

contents one of the things that i was
talking about my mit

presentation and one of the things i
noted is shannon tells us

signal versus noise but it it

i think but it doesn't tell us if the
signal is

a cancer cure or a chocolate chip cookie
recipe

right and so the content is left
undefined

there now what if so

so i define living systems as auto
catalytic

entropy secreting structures

notice i don't say they have to be

embodied in any particular embodiment my

son said the only reason i came up with

the definition is i wanted to use the

word secretion

because it was juicy but he's right in
part
i wanted to make it juicy but the reason
i raised that here
is when you look at barrett-center or
whoever was it barons in any way the
difference that makes a difference
um info information really maybe
is what we cognitive beings call
relevant differentials that we perceive
irrelevant differentials are just noise
and so the difference that makes a
difference we call it information but
there's a lot of differentials out there
in the world
and that's the reason i'm going there is
since we're talking to a philosopher
today
and what i deal with as a lawyer is
rhetoric
and rhetoric is what you do to bring
together incommensurables
right and rhetoric is not like crazy
glue it's like
it's like um epoxy it doesn't just bring
things together it also brings them
together and fills gaps
so the rhetoric of information
is our storytelling about
relevant differentials and we seek to
sculpt our internal model and then train
other people's internal models
because when we entrain other people's
internal models with our rhetoric
then it de-risks our externality because
other people's internal models are my
externality right so the rhetoric
is the tool that we use to reduce

differentials or to increase relevance
alignment among cognitive systems
running cognition how's that i wanted to
try to integrate
it that seemed to me that last couple of
comments
it felt like it may be information is an
article when we put
much in seeking information maybe
information the difference that makes a
difference
after the fact we can tell that because
before the fact we can't tell
what's relevant for instance suntan
lotion never existed because we didn't
know that photons caused skin cancer
now there's an entire commercial
factories and areas in the grocery store
that sell suntan lotion
now it doesn't mean that the sun became
carcinogenic
we just learned that it did so the
inference we were able to make that
inference at large scales
and now influence buying behavior which
leverages and de-risks future behaviors
which is the
reason that we care about information
that's the difference that makes a
difference anyways
a couple of thoughts on this last couple
of things that's that's really
cool stuff i mean i i really see things
similarly i think on a lot of fronts
there
um i think uh
uh so the question of it sounded
to me like you're kind of saying like we

don't have to be too
fixated on this term information because
it really it means
differences that make a difference or
being aware of differentials
so i mean i i
agree with that i think um i mean i i
again i always go back to just the
the sort of bedrock stuff so like i
think
i think shannon's interpretation of
information was actually a little bit
unnecessarily epistemic
i think it could have been just doxastic
so he says you know
it's coming to know something it's like
well
but we could generalize that and just
talk about what one comes to believe
and then i think if we have if we work
with that notion which isn't information
anymore
as shannon defined it because he talked
about knowledge but whatever that
generalization of information is called
um i think that would really um conduct
the logic thanks um yeah so um
so that's that's all that i think we're
getting at with the concept of
information but what what i think it's
really interesting and difficult
philosophically is
okay so obviously there's there's self
information right so there's just like
something pings my retina and or you
know
something downstream from that and i
notice a little flip in my visual field

right

and that's that's something i'm aware of

so that's information about itself

anyway

but then the you know the more important

signals carry all sorts of information

about other things

meaning they i guess they lead us to

predict and expect other differences

beyond themselves and so and so the

tricky thing to me is okay so

i take myself to be this system of

interacting parts

neurons and also other things obviously

that um

sort of embodies a perspective and the

information that a signal carries

depends on

that whole system and how this signal

impacts

it right and and you know one of my

neurons is a

entirely different type of system here's

my daughter

tell me something what's up you got ice

cream awesome

sorry um

that's awesome um yeah so

your generative model's on a live stream

right now i'm sorry

that's cool yeah um i don't know so i

was saying something about neurons and

then now i'm thinking about ice cream

um basically yeah so so the

tr like that the question for me is so

like i think of content right the

information content of a signal for a

person as relative to

the person to the system and so if
you've got a completely different system
with a different structure
i think you can sort of compare the
structures of the systems to see how
similar the content is right but i think
it has to be at that whole level
of the sort of holistic entire system
level that you need to make these
content comparisons
um so i don't remember whether that
directly links back to where we started
this but
um nice
in one one quick thought on that is the
the difference that makes a difference
which is also a big key term in the
philosophy of biology
like the mutation is the difference that
makes a difference for something
um in active inference it's the
difference in
observation that makes a difference for
either the generative model updating
that's inference
or the difference that makes a
difference for action so we have a way
to talk about it and then
also um the doxastic note
not my total area but it looks like this
is how we can work towards pluralism
across perspectives because if it's
framed as one person believes that it's
hot and one person believes that it's
cold
if there's only one truth and we're
gonna play tug of war then
we have to fight but if it's true that

one person believes it's hot and it's
true that one person believes it's cold
there's no problem there still might be
differences on what temperature the
people want the room
but by framing things about what a
perspective or a system or an agent
believes to be true or just believes
period could be an ironic belief
all of a sudden we can interact with
multi-agent systems
and deal with them in a way that still
has local logic
so steven and then anyone else
so thanks just so closing out i suppose
in a way
i'd be curious how this this work with
beliefs that
has a nice as as daniel said it bridges
action orientated approaches and
pragmatic approaches with more
epistemic approaches because beliefs
tend to be beliefs about
what will happen when something is done
in the world
often not necessarily and i'm curious
how this paper and this kind of
underpinning relates or helped to
inform where you were going with a new
paper that was intriguing about the
folk psychology and how maybe that that
there's some sort of that's where one of
the bridges out
from this paper into
you know folk psychology being how we
kind of intuitively believe the world is
based on our experiences so there is a
certain element of like

whether that's how it's structured in
the brain it's kind of how
we might structure it before science
tells us something else
in a in a certain socio-cultural sense
so i'd be interested in
how that extension plays out with this
paper
yeah so so there's definitely a direct
connection between this paper and
and that one i mean i wanted to write a
sort of sequel to this paper where i
talked about
specifically desires and how those the
the sort of
the way those would be grounded in
thermodynamics according to this picture
or co-native states in general like the
states with a world to mind direction of
fit um
and so then i started talking to to ryan
smith and others about this and we ended
up
you know co-authoring a quite different
paper that
has some commonality but i still want to
write um
i still want to write one that's more on
the thermodynamic side of things but
um the connection is in this you know
the sort of identity thesis
that at the end of this paper i argue
for it by way of david lewis who
who one of the terms in the identity is
the implicit definition
that folk psychology gives us of mental
states
um so we haven't really gotten into that

maybe we will next week or something i
don't know there's ramsey sentences
i'd love to talk about them we got we
got the ramsey sentences
we got the sleep wake simulation yeah
other topics that we will really look
forward to exploring
yeah let's please get into that but um
like
basically i mean i don't know if as
usual
i feel i always feel like i'm not doing
an adequate job of answering questions
i'm sorry
but um but by folk psychology in this
context i'm really thinking of just
our our it's sort of like theory of mind
right it's like our
implicit theory of the way people
including ourselves
um function so i think that
that's essential even though it's not
highly until the end that's essential to
what i'm doing in this paper because if
you don't have
mental states to define then you don't
have any identity
relation because that's one of the terms
in the identity relation although you
could also i guess
maybe if you're doing pure bayesian
modeling you could just write down a
bunch of states and
equations for how they update um without
thinking about spoke psychology at all
but but i guess the point of this is
that
um if one of the terms in this relation

is the mind
well i think folk psychology i don't
actually love that term i mean
it's okay it's a philosopher's term of
art right i just mean
intuitive human psychology i think is
one of the terms in this relation
cool scientists are folk too so
cognitive diversity means that we have
all different ways of knowing and there
may not be one
default or simply intuitive way of
working or knowing because what's
intuitive to one person
will be out of the groove for somebody
else and that's why the
movement towards pluralism in terms of
our understanding of
ideas and of ecosystems is so
interesting to talk about and see
connected to active inference
any um thoughts or what would be
something that somebody's excited
from the conversation today or thinking
about how we could resolve
or at least introduce again next week
i think a few of the things we have
written down is the sleep wake
simulation what is that agent doing what
does it do in terms of your model
we have the ramsay sentences and where
does that
style of logical representation come
into play we have
empirical validity what kind of
experiments
are you interested in or curious about
what what else could be interesting

uh we could still talk about the the
contrast of divergence versus simulated
annealing thing
um i just really briefly wanted to
follow up and say i
think when i say intuitive psychology
i mean i don't mean to suggest that
there's just one of those i think that
but i do think we can sort of average
over different perspectives to get
that so like uh the individual
perspectives ultimately are what's real
but if we want to talk about things at
an abstract level we could average over
those and i think that's kind of what
lewis tries to do so
maybe this is something we'll get into
next week um but
really he when he talks about folk
psychology
he he he engineers it so that it doesn't
have to be that there's just one set of
state statements that embodies what
people think about psychology it's like
a disjunction of different um sets that
mostly overlap
um um and just yeah i mean
pluralism is really i think is um
is a good idea in many contexts um i
actually think
this would be interesting this might be
controversial i think it's not always
the goal in science because if you start
out with pluralism
then you just don't do anything right
it's like okay
everything's fine as it is but i think
sometimes we're seeking a unifying

underlying explanation and if you accept
pluralism at the start then you won't
get there
that tension which is okay pluralism
everybody must be pluralist and this is
the only way
it's actually absolutist and singular at
a higher level
and so pluralism means that we each do
have our perspectives and yet there is a
way of working
it's not just the same thing as saying
we have an absolute answer
and it's that there's no answer that's
the kind of local logical paradoxes that
we see all around us
that's a great point stephen and then
anyone else
yes that's that's the um the rabbit hole
of post-modernism
oh we don't want to go but it's it's
it's good
this is a really rich conversation and
uh i think like you say it's uh
to be able to jump from helmholtz to
some of these trans-cultural things is
uh you know it shows you that it's
pretty foundational some of the stuff
we're touching so that's great
so thanks for a lovely a lovely talk
today i really enjoyed it thank you
yeah thanks a lot any closing notes
there alex otherwise we're all gonna
probably
re-listen and re-read and think about
what would be good to ask next week
yeah i know thank thank you all it's
it's really fascinating and

fun and i just want to do more of it
next week i'm
i'm very uh i'm
i didn't have anything else to say but i
kept speaking i'm not sure why
same way do that do that sometimes it
happens
so thanks again everybody who's
participating live and on live stream
and we will see you next week